Going nonparametric:

Nearest neighbor and kernel regression

Emily Fox & Carlos Guestrin

Machine Learning Specialization

University of Washington

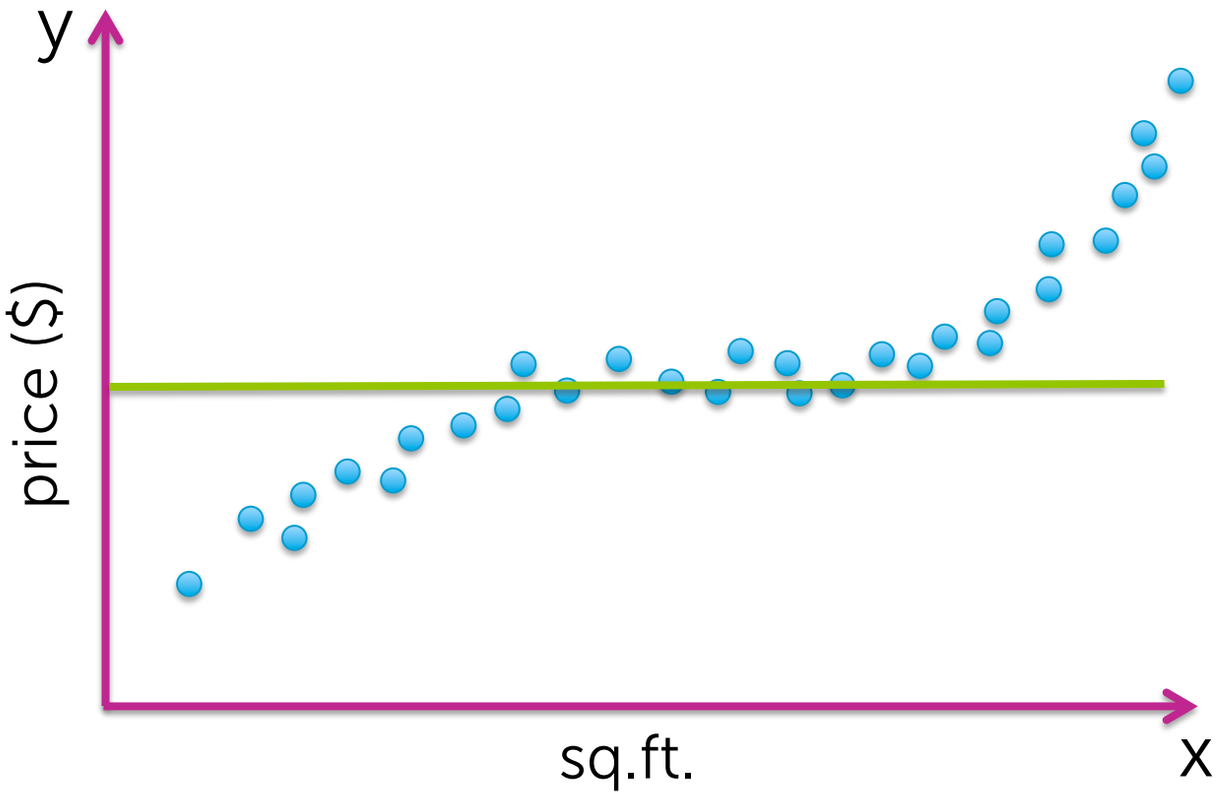
©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Fit globally vs. fit locally

©2015 Emily Fox & Carlos Guestrin

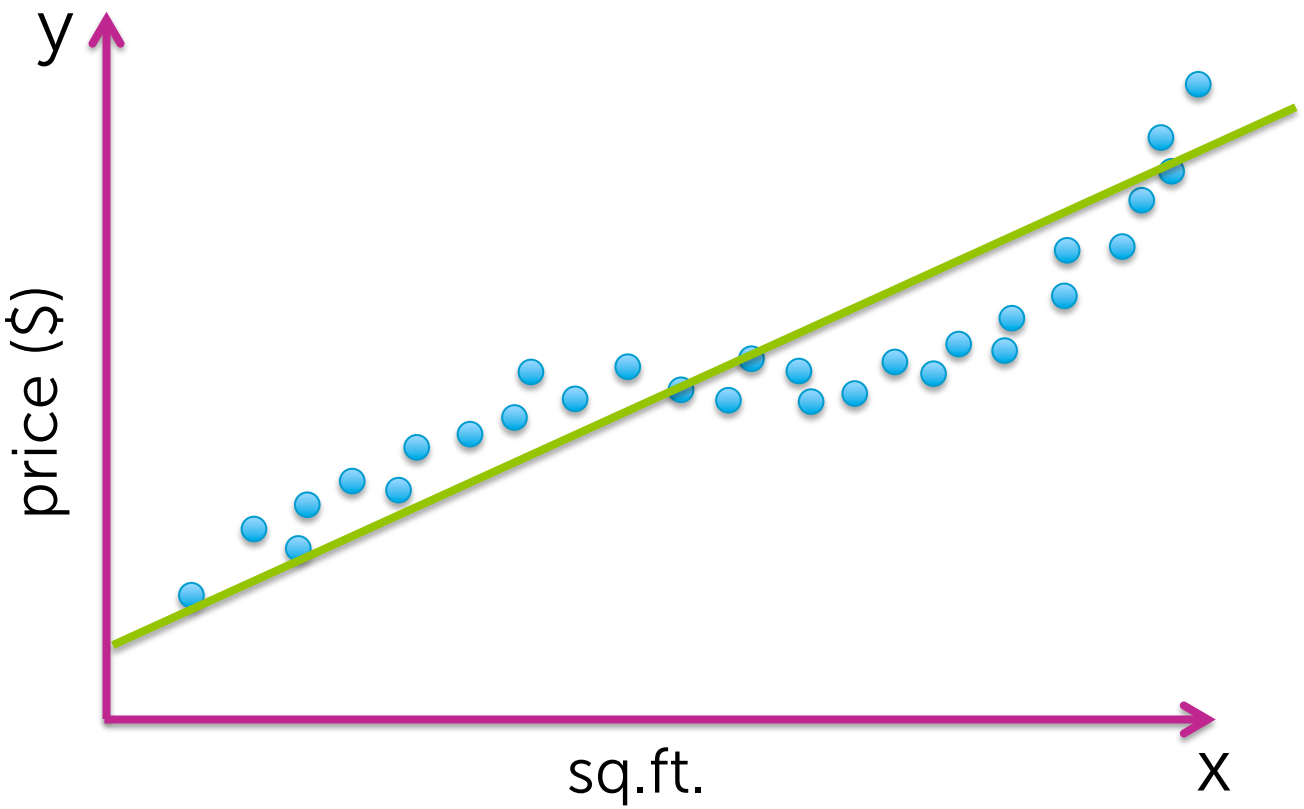
Parametric models of f(x)



©2015 Emily Fox & Carlos Guestrin

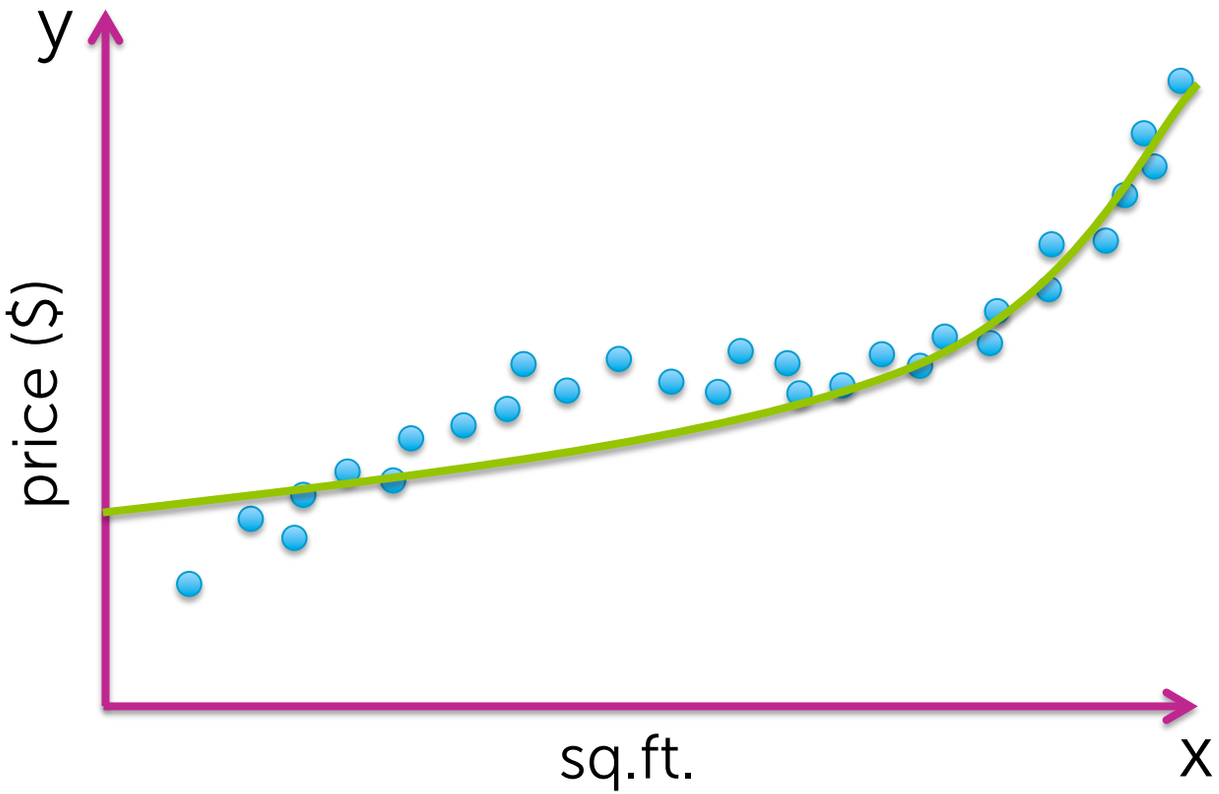
Troy University Resources

Parametric models of f(x)



©2015 Emily Fox & Carlos Guestrin

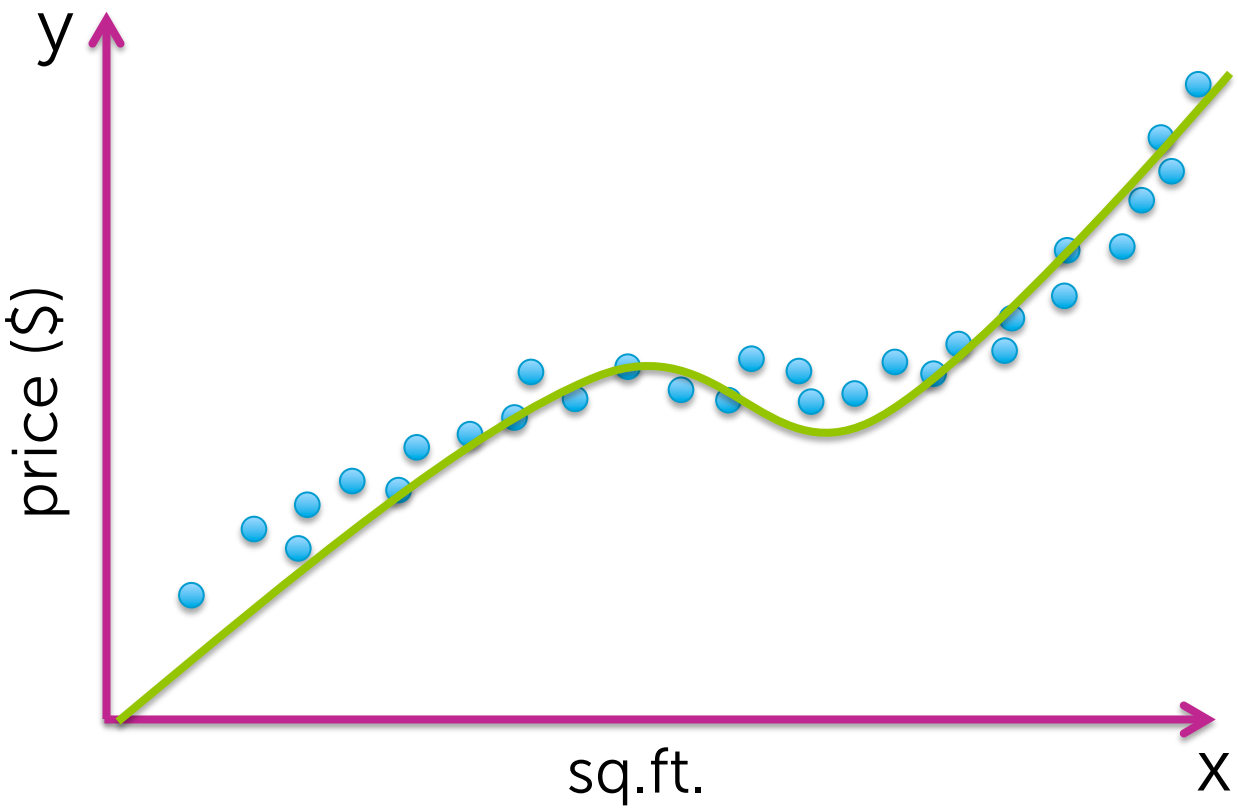
Parametric models of f(x)



©2015 Emily Fox & Carlos Guestrin

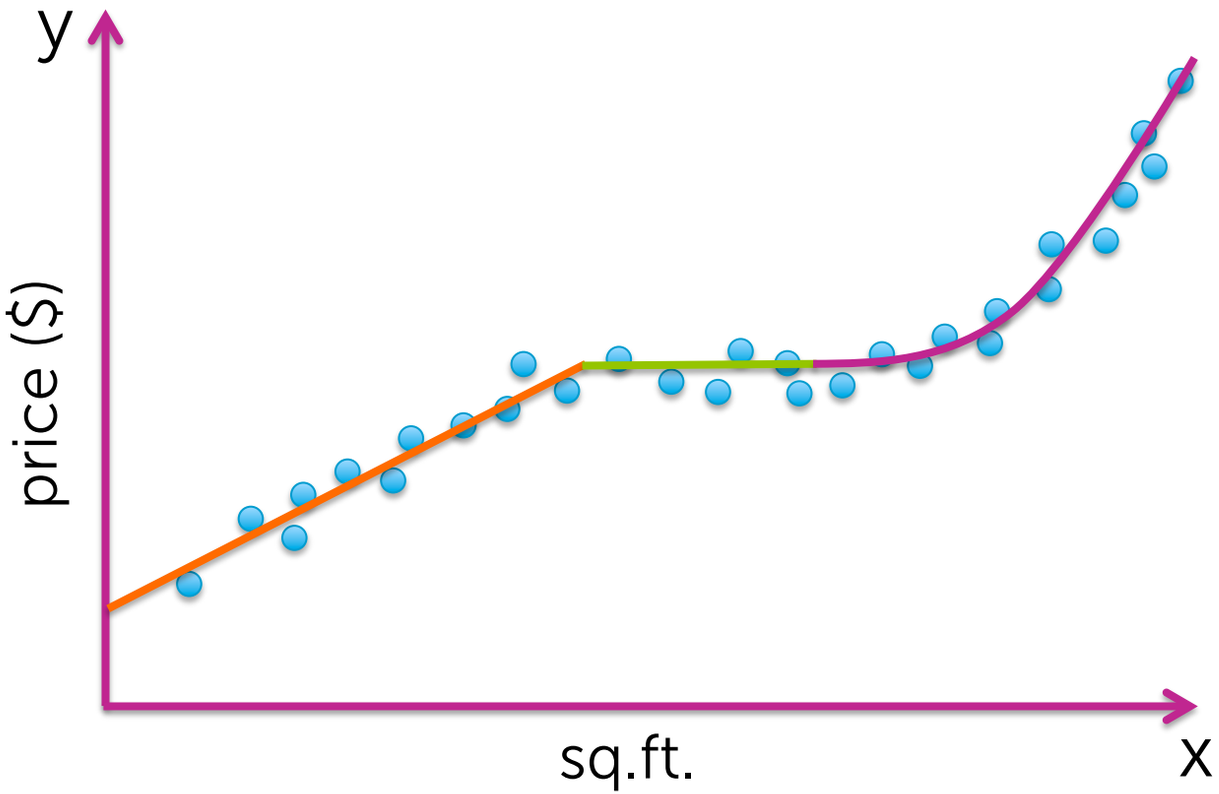
Troy University Resources

Parametric models of f(x)



©2015 Emily Fox & Carlos Guestrin

f(x) is not really a polynomial



©2015 Emily Fox & Carlos Guestrin

Troy University Resources

What alternative do we have?

If we:

- Want to allow flexibility in f(**x**) having local structure
- Don't want to infer "structural breaks"

What's a simple option we have?

- Assuming we have plenty of data…

©2015 Emily Fox & Carlos Guestrin

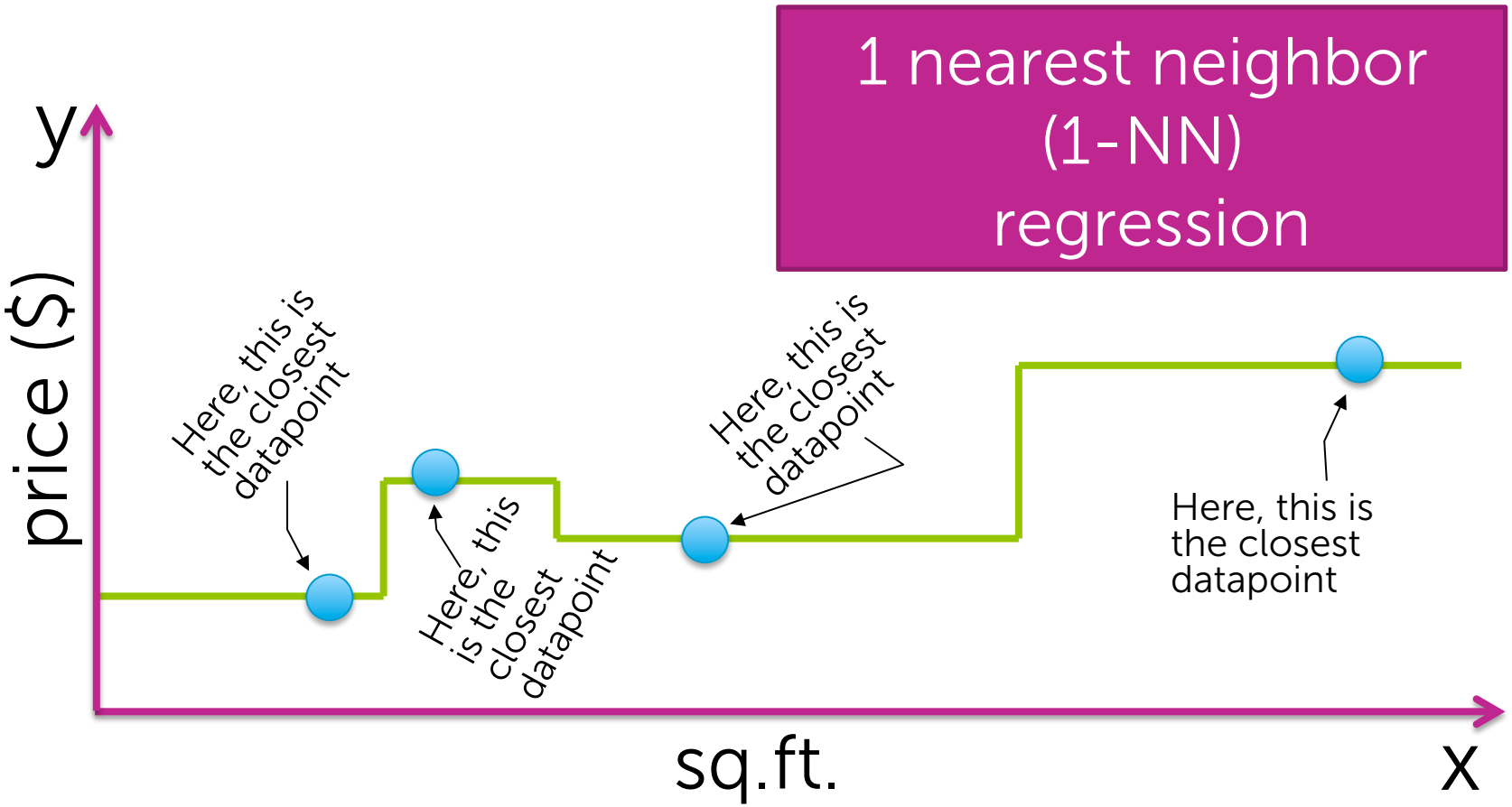
Simplest approach: Nearest neighbor regression

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Fit locally to each data point

Predicted value = "closest" y_i



©2015 Emily Fox & Carlos Guestrin

What people do naturally…

Real estate agent assesses value by finding sale of most similar house



©2015 Emily Fox & Carlos Guestrin

Troy University Resources

1-NN regression more formally

Dataset of (🏠,$) pairs: $(\mathbf{x}_1,y_1), (\mathbf{x}_2,y_2),\ldots,(\mathbf{x}_N,y_N)$

Query point: $\mathbf{x}_q$ ←  $?

big lime green house

1. Find "closest" $\mathbf{x}_i$ in dataset

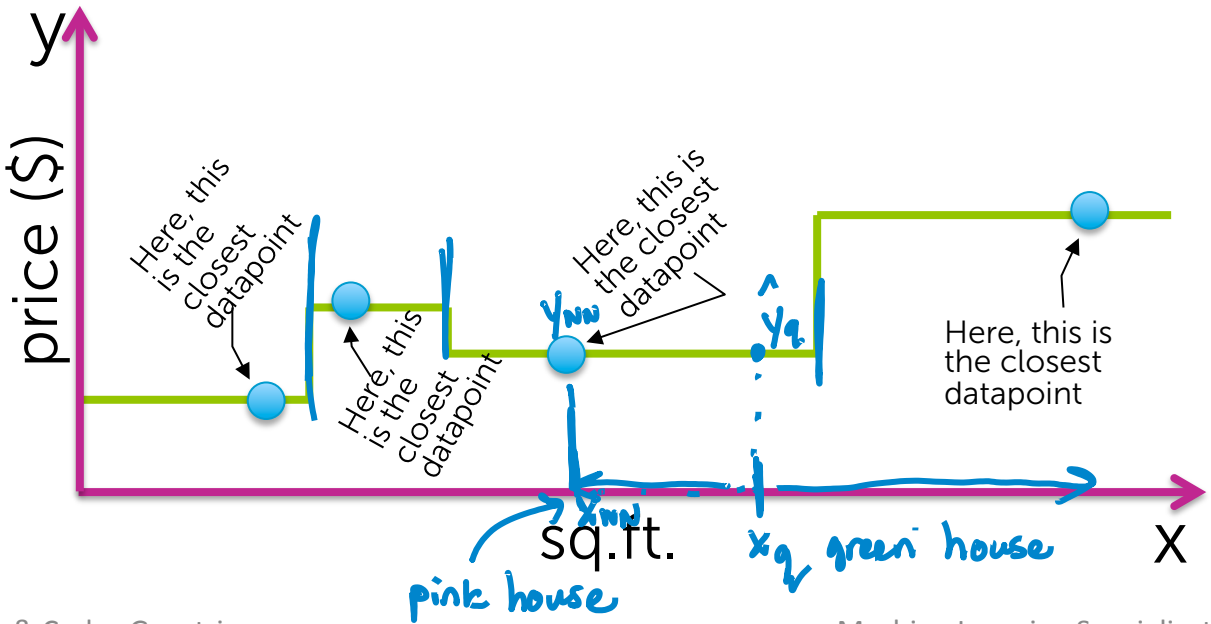
$x_{NN} \leftarrow \min_i \text{distance}(x_i, x_q)$

big pink house

2. Predict

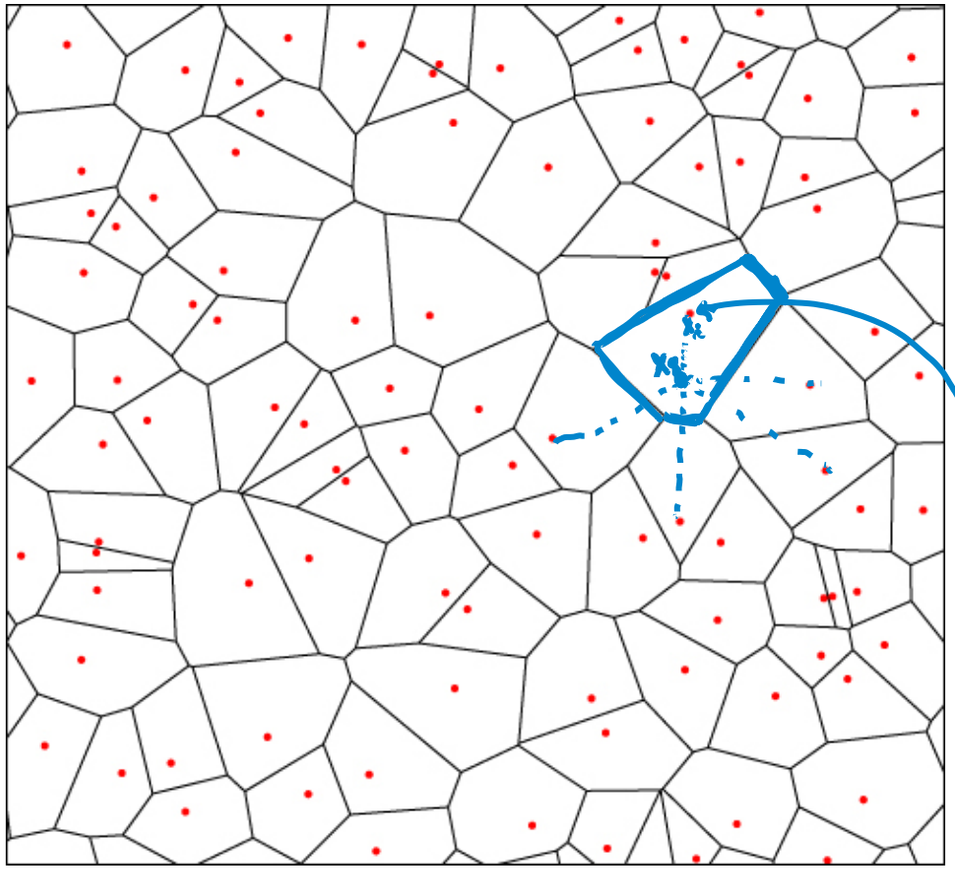
$\hat{y}_q = y_{NN}$

sales price of big pink house



©2015 Emily Fox & Carlos Guestrin

Visualizing 1-NN in multiple dimensions



Voronoi tesselation (or diagram):

- Divide space into N regions, each containing 1 datapoint
- Defined such that any **x** in region is "closest" to region's datapoint

x_q closer to x_i than any other x_j for $j \neq i$.

Don't explicitly form!

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Distance metrics: Defining notion of "closest"

In 1D, just Euclidean distance:

$$\text{distance}(x_j, x_q) = |x_j - x_q|$$

In multiple dimensions:

- can define many interesting distance functions
- most straightforwardly, might want to weight different dimensions differently

©2015 Emily Fox & Carlos Guestrin

Weighting housing inputs

Some inputs are more relevant than others



bedrooms
bathrooms
sq.ft. living
sq.ft. lot
floors
year built
year renovated
waterfront



©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Scaled Euclidean distance

Formally, this is achieved via

$$\text{distance}(\mathbf{x}_j, \mathbf{x}_q) = \sqrt{a_1(\mathbf{x}_j[1]-\mathbf{x}_q[1])^2 + \ldots + a_d(\mathbf{x}_j[d]-\mathbf{x}_q[d])^2}$$

weight on each input
(defining relative importance)

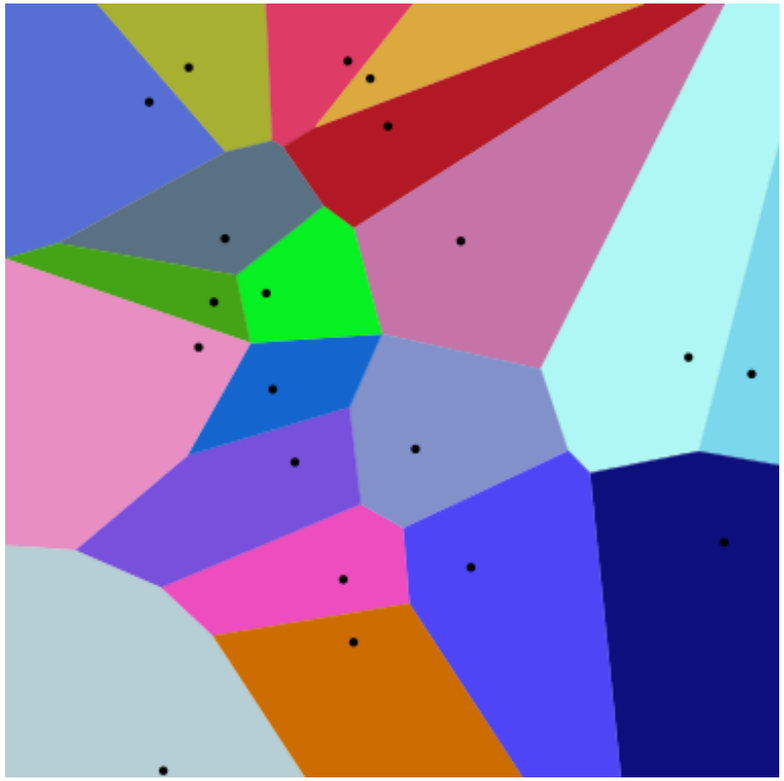
Other example distance metrics:

– Mahalanobis, rank-based, correlation-based, cosine similarity, Manhattan, Hamming, …

©2015 Emily Fox & Carlos Guestrin

Different distance metrics lead to different predictive surfaces

Euclidean distance



Manhattan distance



©2015 Emily Fox & Carlos Guestrin

Troy University Resources

1-NN algorithm

©2015 Emily Fox & Carlos Guestrin

Performing 1-NN search

- Query house:



- Dataset:



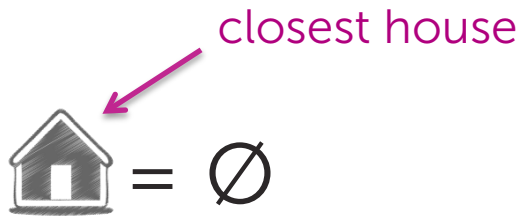
- **Specify:** Distance metric
- **Output:** Most similar house



©2015 Emily Fox & Carlos Guestrin

Troy University Resources

1-NN algorithm



Initialize **Dist2NN** = ∞, = ∅

For i=1,2,…,N



Compute: δ = **distance(** $_i$, $_q$ **)**

If δ < **Dist2NN**

set $_i$

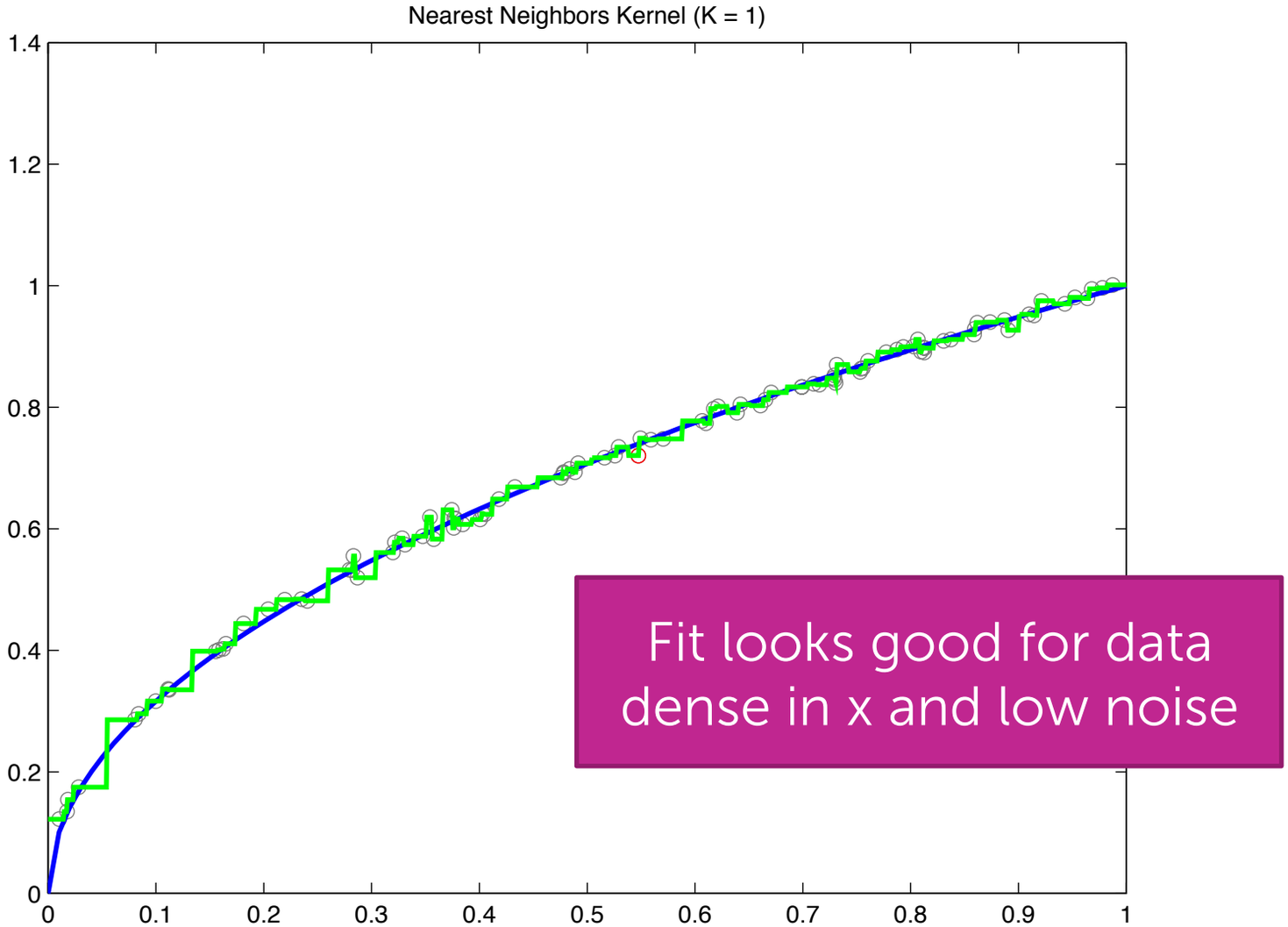
set **Dist2NN** = δ

Return most similar house



©2015 Emily Fox & Carlos Guestrin

1-NN in practice

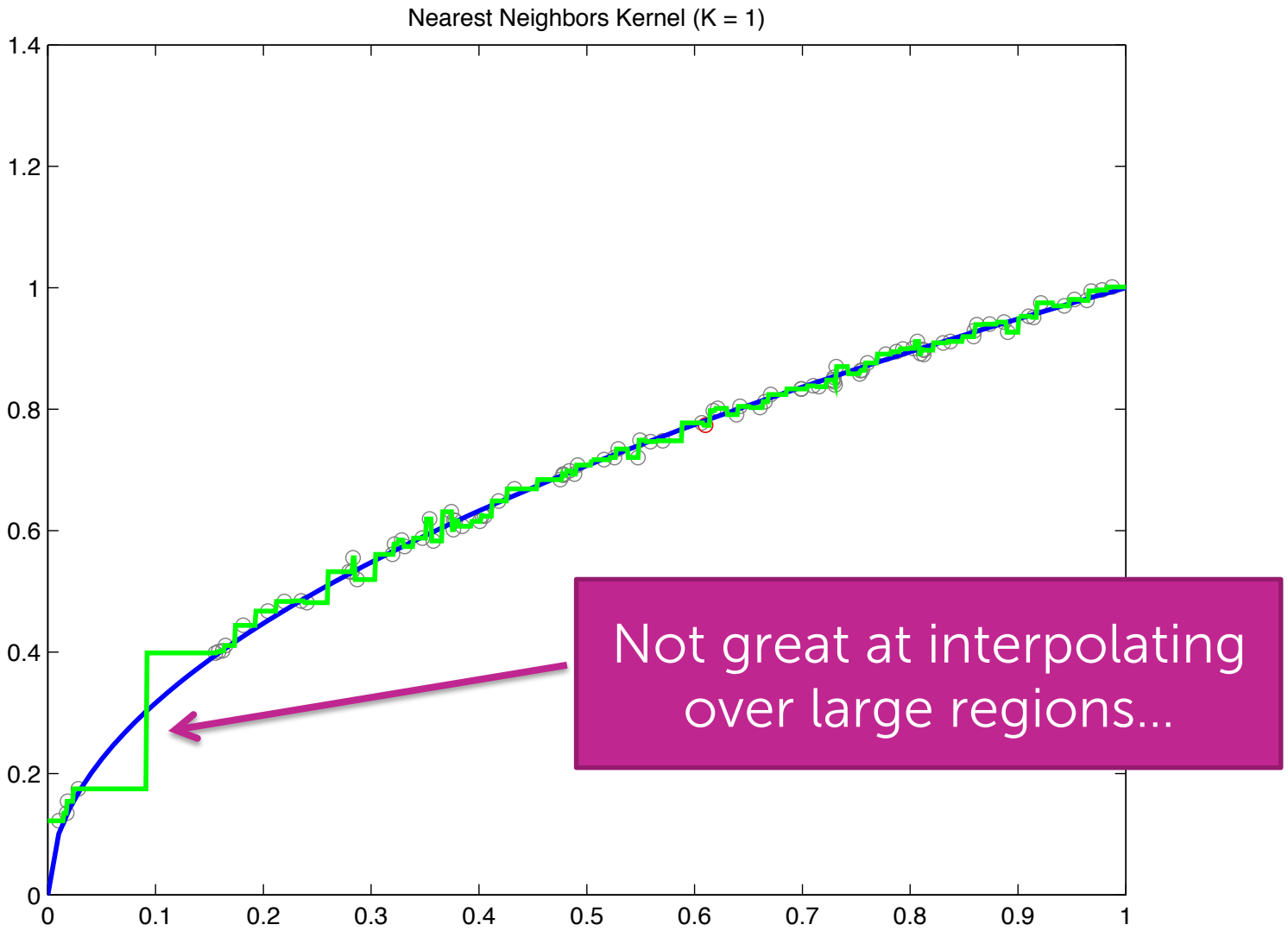


Fit looks good for data dense in x and low noise

©2015 Emily Fox & Carlos Guestrin

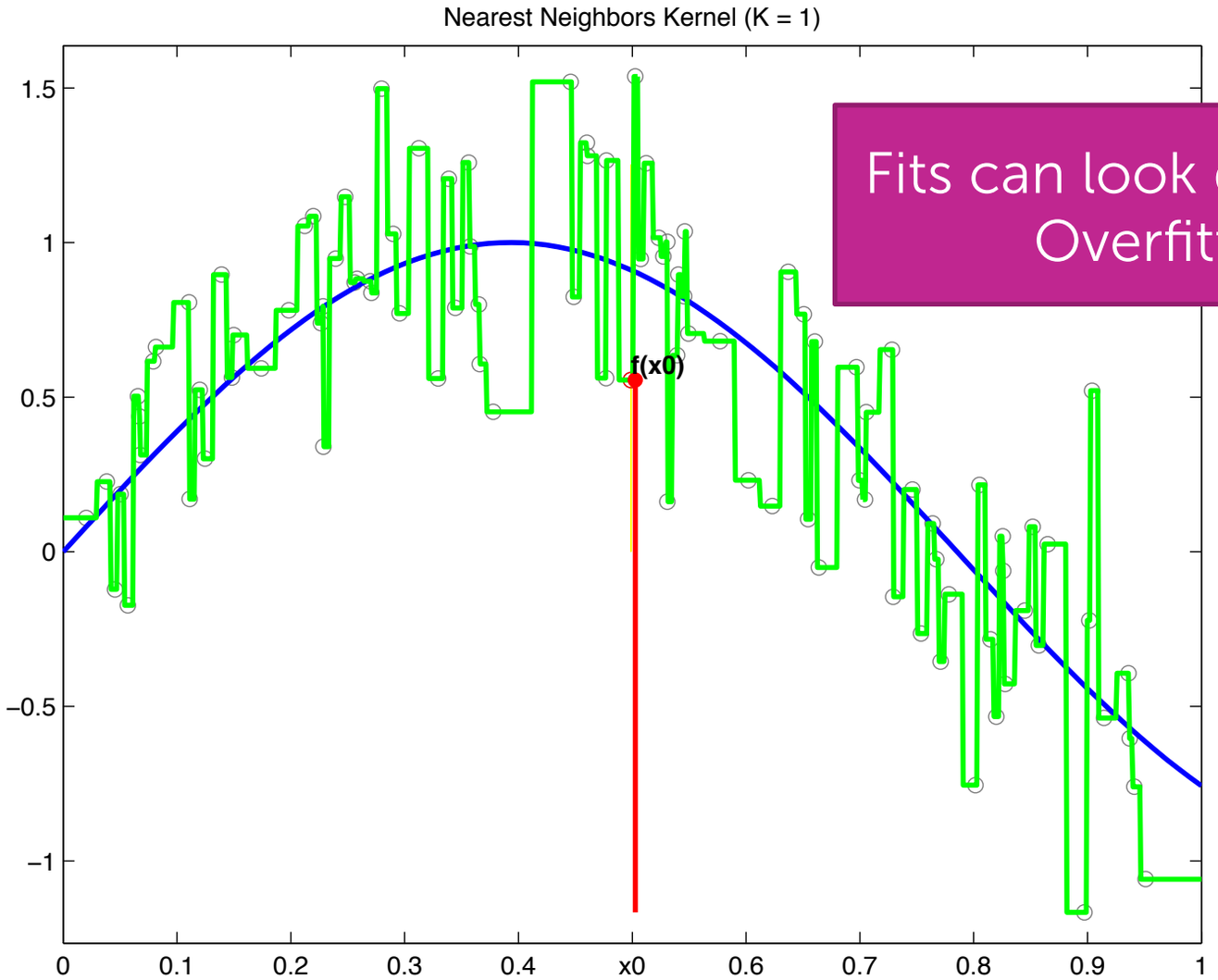
Troy University Resources

Sensitive to regions with little data



©2015 Emily Fox & Carlos Guestrin

Also sensitive to noise in data



Fits can look quite wild… Overfitting?

©2015 Emily Fox & Carlos Guestrin

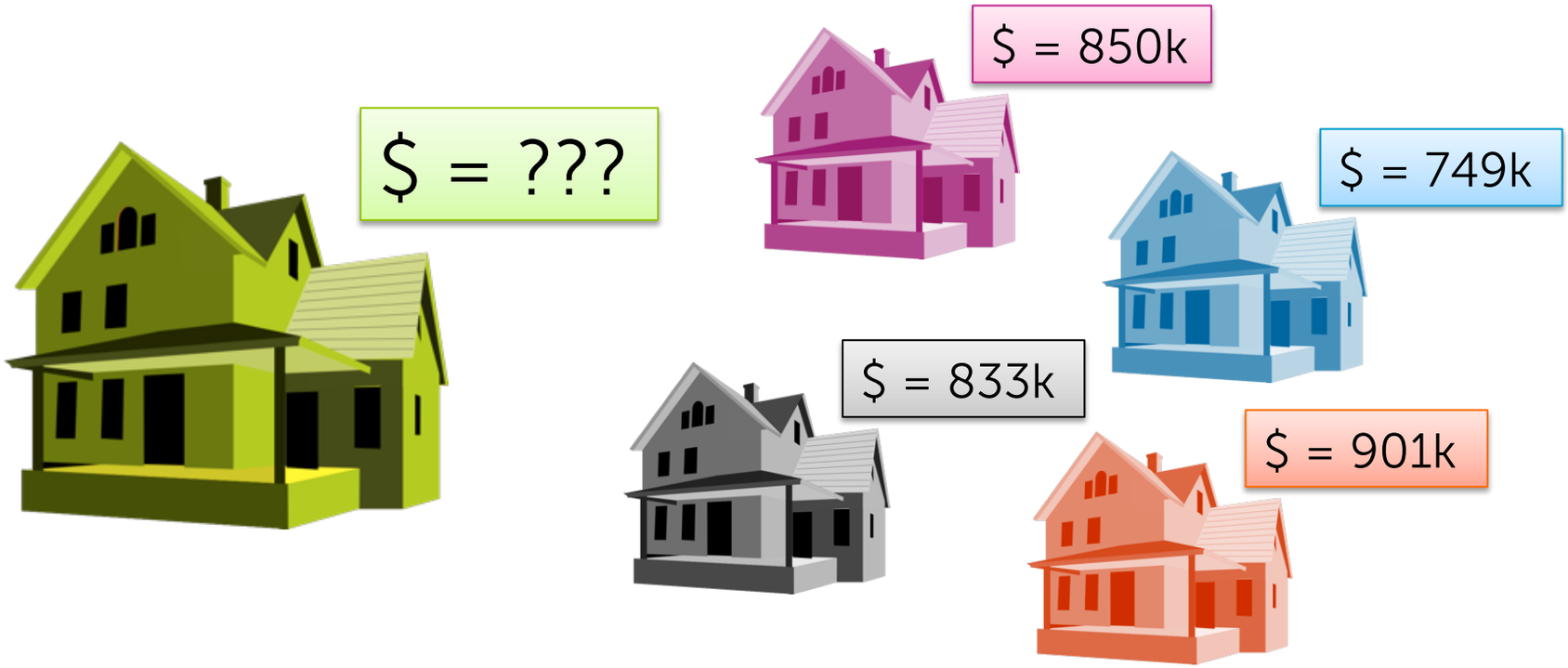
Troy University Resources

k-Nearest neighbors

©2015 Emily Fox & Carlos Guestrin

Get more "comps"

More reliable estimate if you base estimate off of a larger set of comparable homes



©2015 Emily Fox & Carlos Guestrin

Troy University Resources

k-NN regression more formally

Dataset of ( ,$) pairs: $(\mathbf{x}_1,y_1)$, $(\mathbf{x}_2,y_2)$,…,$(\mathbf{x}_N,y_N)$

Query point: $\mathbf{x}_q$

1. Find k closest $\mathbf{x}_i$ in dataset

$(x_{NN_1}, x_{NN_2}, \ldots, x_{NN_k})$ such that for any x_i not in nearest neighbor set,

$$\text{distance}(x_i, x_q) \geq \text{distance}(x_{NN_k}, x_q)$$

2. Predict

$$\hat{y}_q = \frac{1}{k}(y_{NN_1} + y_{NN_2} + \ldots + y_{NN_k})$$

$$= \frac{1}{k}\sum_{j=1}^{k} y_{NN_j}$$

©2015 Emily Fox & Carlos Guestrin

Performing k-NN search

- Query house:



- Dataset:



- **Specify:** Distance metric
- **Output:** Most similar houses



©2015 Emily Fox & Carlos Guestrin

Troy University Resources

k-NN algorithm

sort first **k houses** by distance to query house

Initialize **Dist2kNN** = sort($\delta_1,\ldots,\delta_k$) ← list of sorted distances

= sor[house] ,…, [house]$_1$ [house]$_k$ ← list of sorted houses

For i=k+1,…,N

query house

Compute: δ = **distance(**[house]$_i$, [house]$_q$**)** [house]

If δ < **Dist2kNN**[k]

find j such that δ > **Dist2kNN**[j-1] but δ < **Dist2kNN**[j]

remove furthest house and shift queue:

[j+1] = [j:k-1]

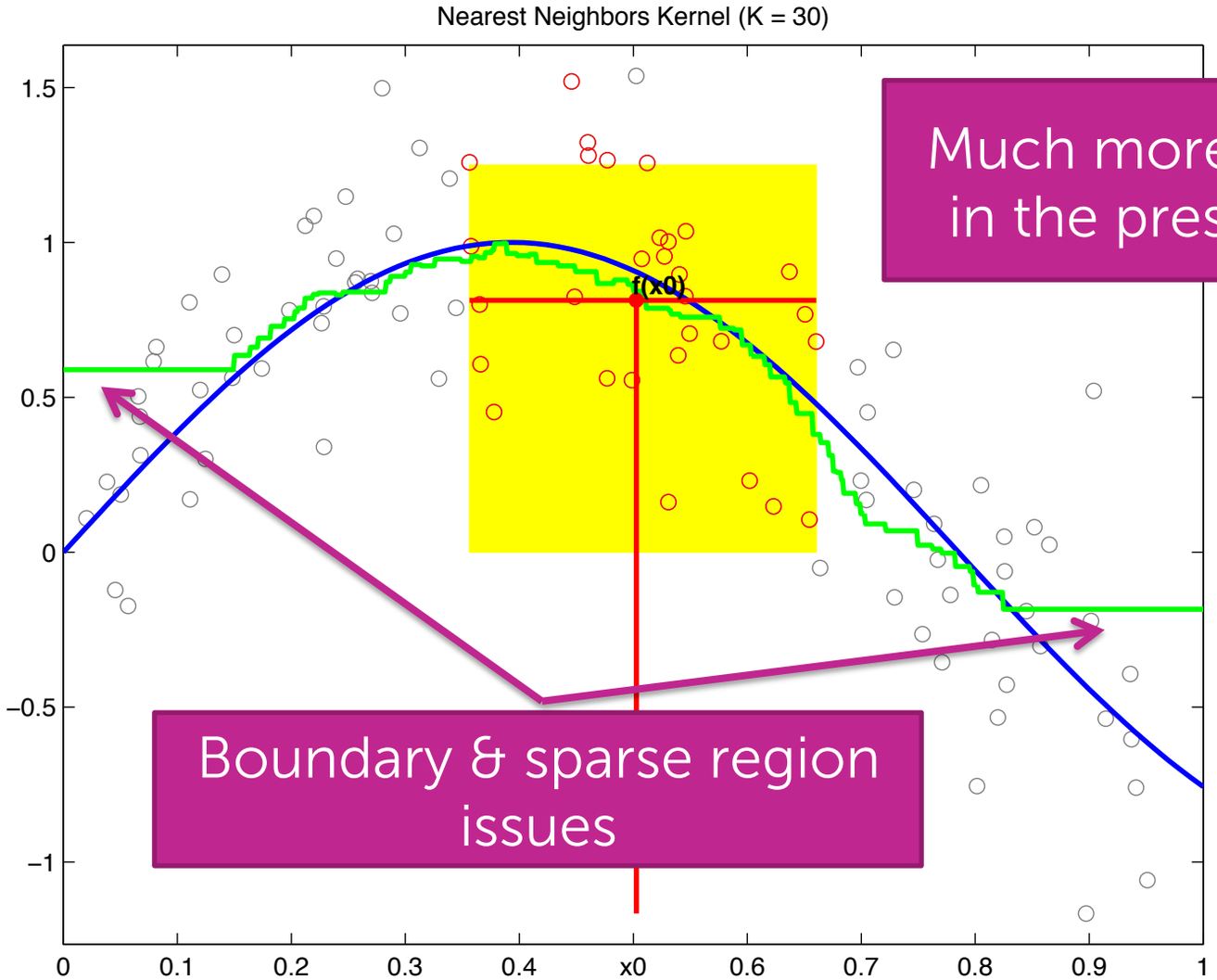
Dist2kNN[j+1:k] = **Dist2kNN**[j:k-1]

insert NN new

set **Dist2kNN**[j] = δ and [house] = [house]$_i$

Return k most similar houses ← closest houses to query house [houses]

k-NN in practice



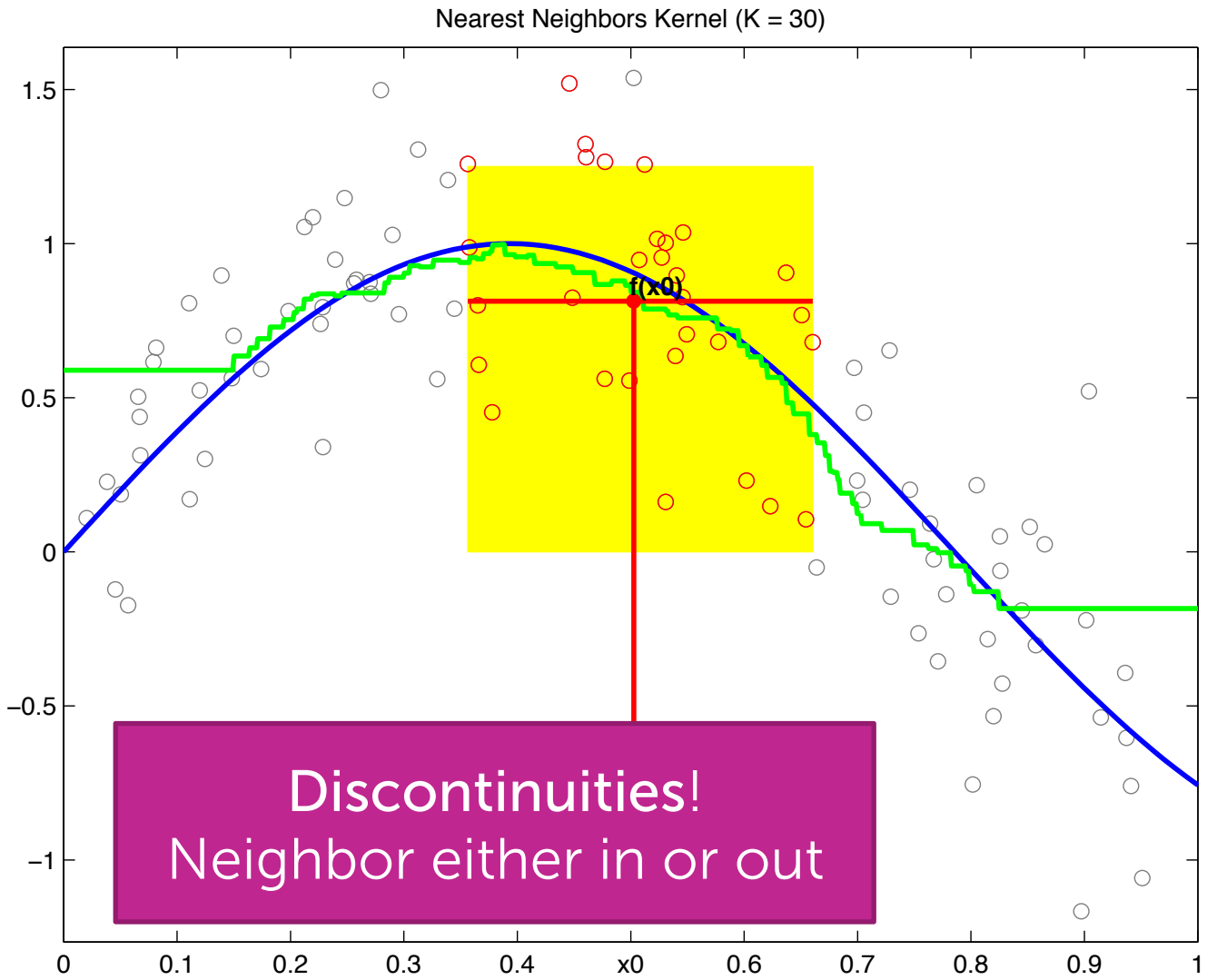
Much more reasonable fit in the presence of noise

Boundary & sparse region issues

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

k-NN in practice



©2015 Emily Fox & Carlos Guestrin

Issues with discontinuities

Overall predictive accuracy might be okay, but…

For example, in housing application:

- If you are a buyer or seller, this matters
- Can be a jump in estimated value of house going just from 2640 sq.ft. to 2641 sq.ft.
- Don't really believe this type of fit

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

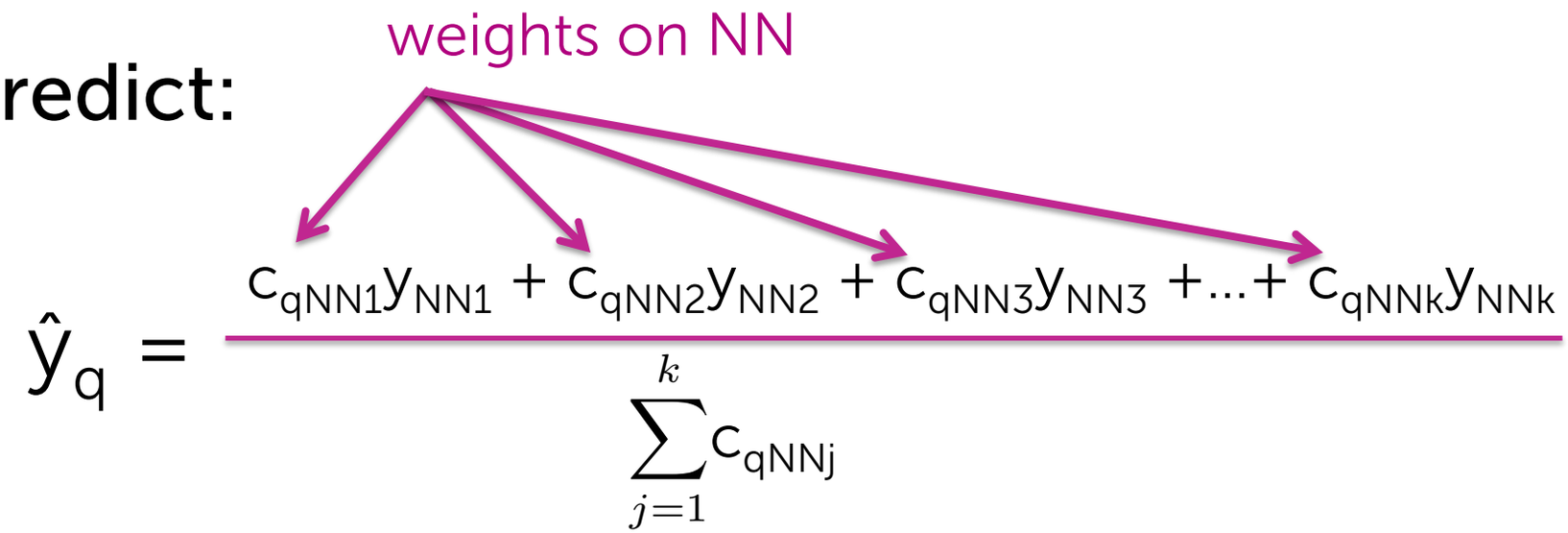
Weighted k-nearest neighbors

©2015 Emily Fox & Carlos Guestrin

Weighted k-NN

Weigh more similar houses more than those less similar in list of k-NN

Predict:



©2015 Emily Fox & Carlos Guestrin

Troy University Resources

How to define weights?

Want weight c_{qNNj} to be small when

distance($\mathbf{x}_{NNj}$,$\mathbf{x}_q$) large

and c_{qNNj} to be large when

distance($\mathbf{x}_{NNj}$,$\mathbf{x}_q$) small

Simple method:

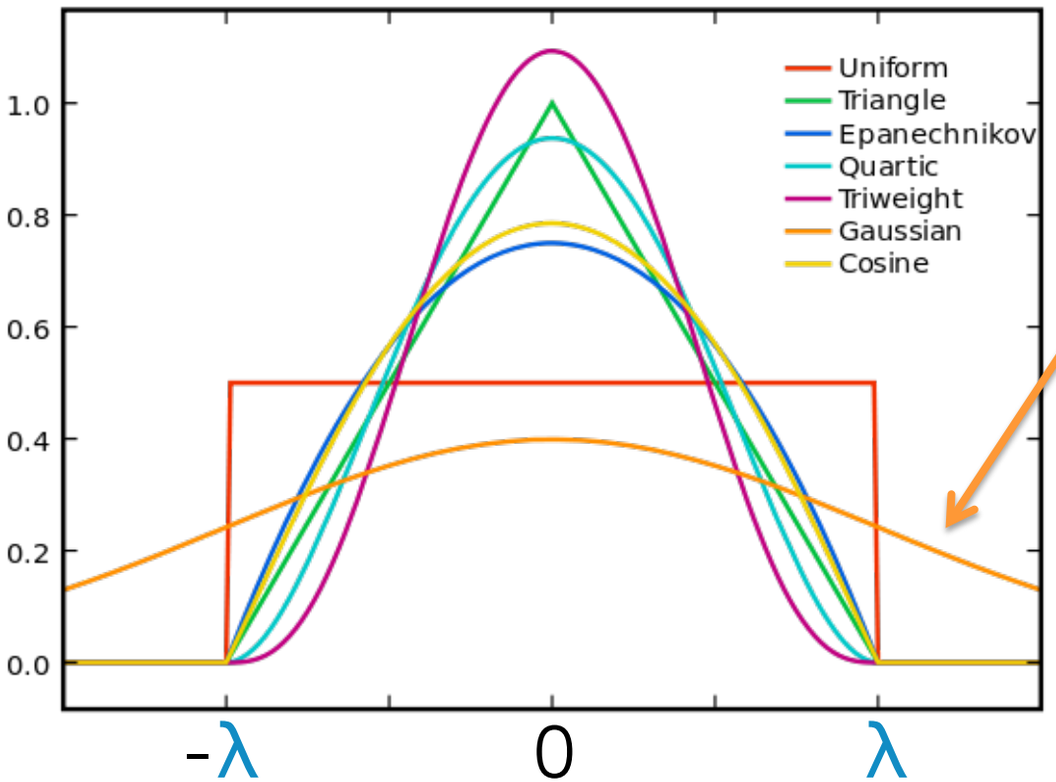
$$c_{q\,NN_j} = \frac{1}{\text{distance}(x_j, x_q)}$$

©2015 Emily Fox & Carlos Guestrin

Kernel weights for d=1

Define: $c_{qNNj} = \text{Kernel}_\lambda(|x_{NNj}-x_q|)$

simple isotropic case



Gaussian kernel:

$\text{Kernel}_\lambda(|x_i-x_q|) = \exp(-(x_i-x_q)^2/\lambda)$

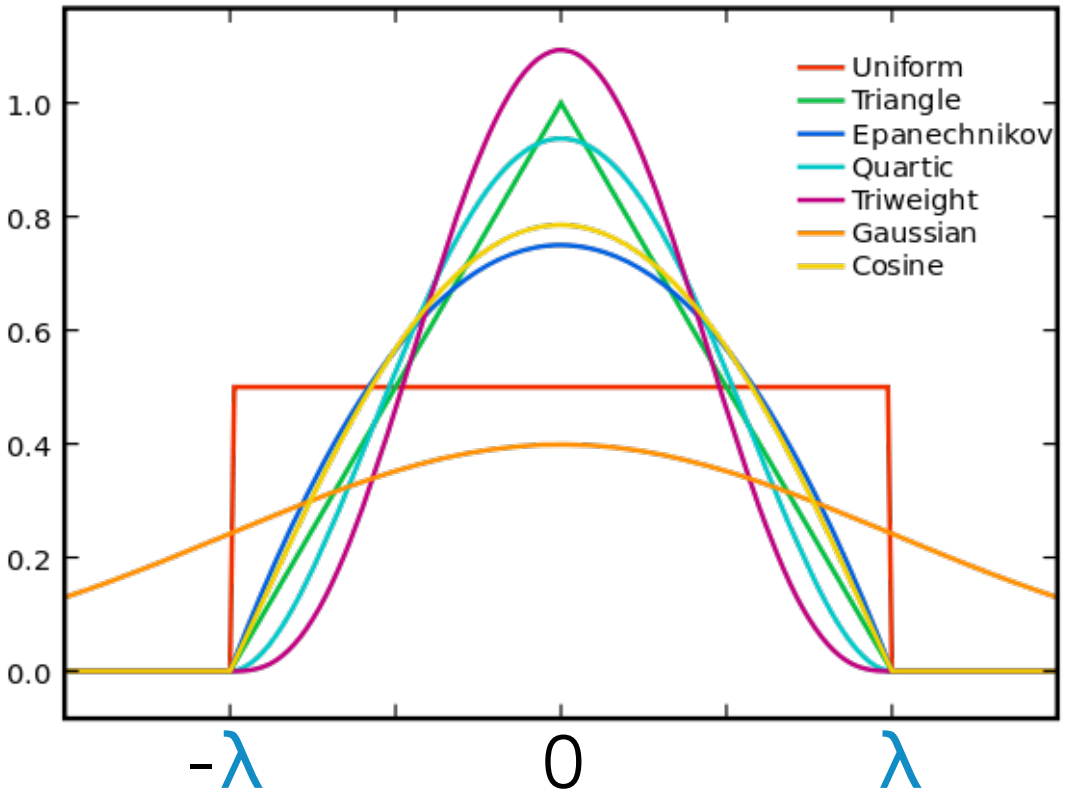
Note: never exactly 0!

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Kernel weights for d≥1

Define: $c_{qNNj} = \text{Kernel}_{\lambda}(\text{distance}(\mathbf{x}_{NNj}, \mathbf{x}_q))$



©2015 Emily Fox & Carlos Guestrin

Kernel regression

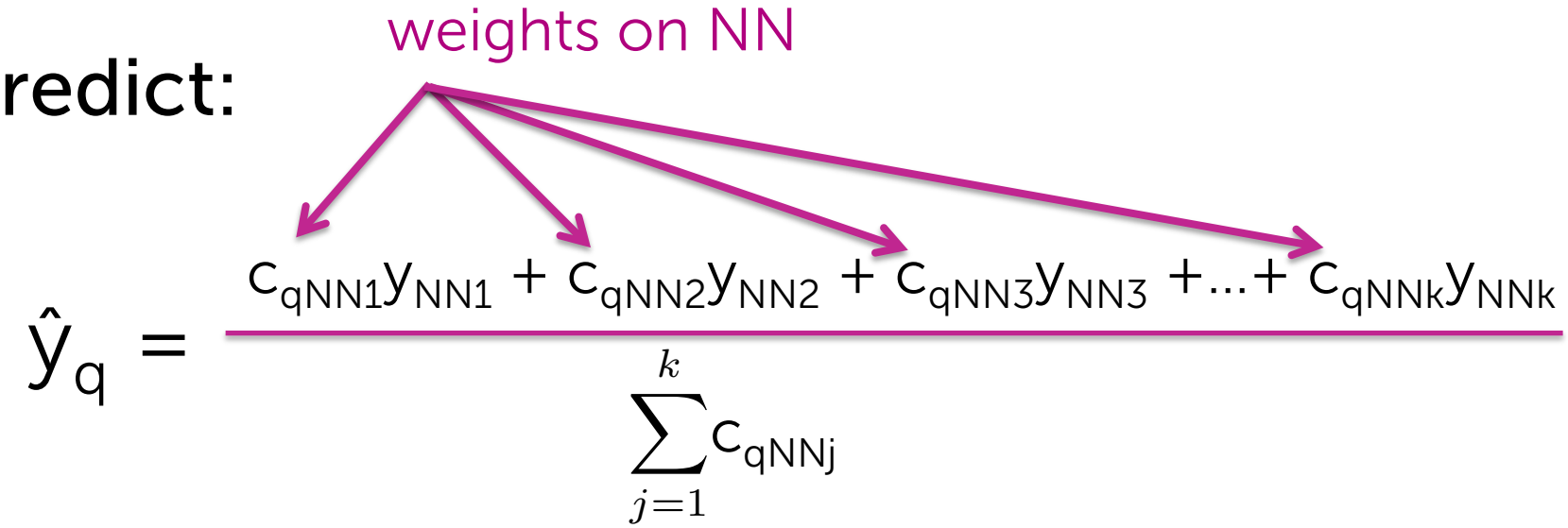
©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Weighted k-NN

Weigh more similar houses more than those less similar in list of k-NN

Predict:



©2015 Emily Fox & Carlos Guestrin

Kernel regression

Nadaraya-Watson kernel weighted average

Instead of just weighting NN, weight *all* points

Predict:

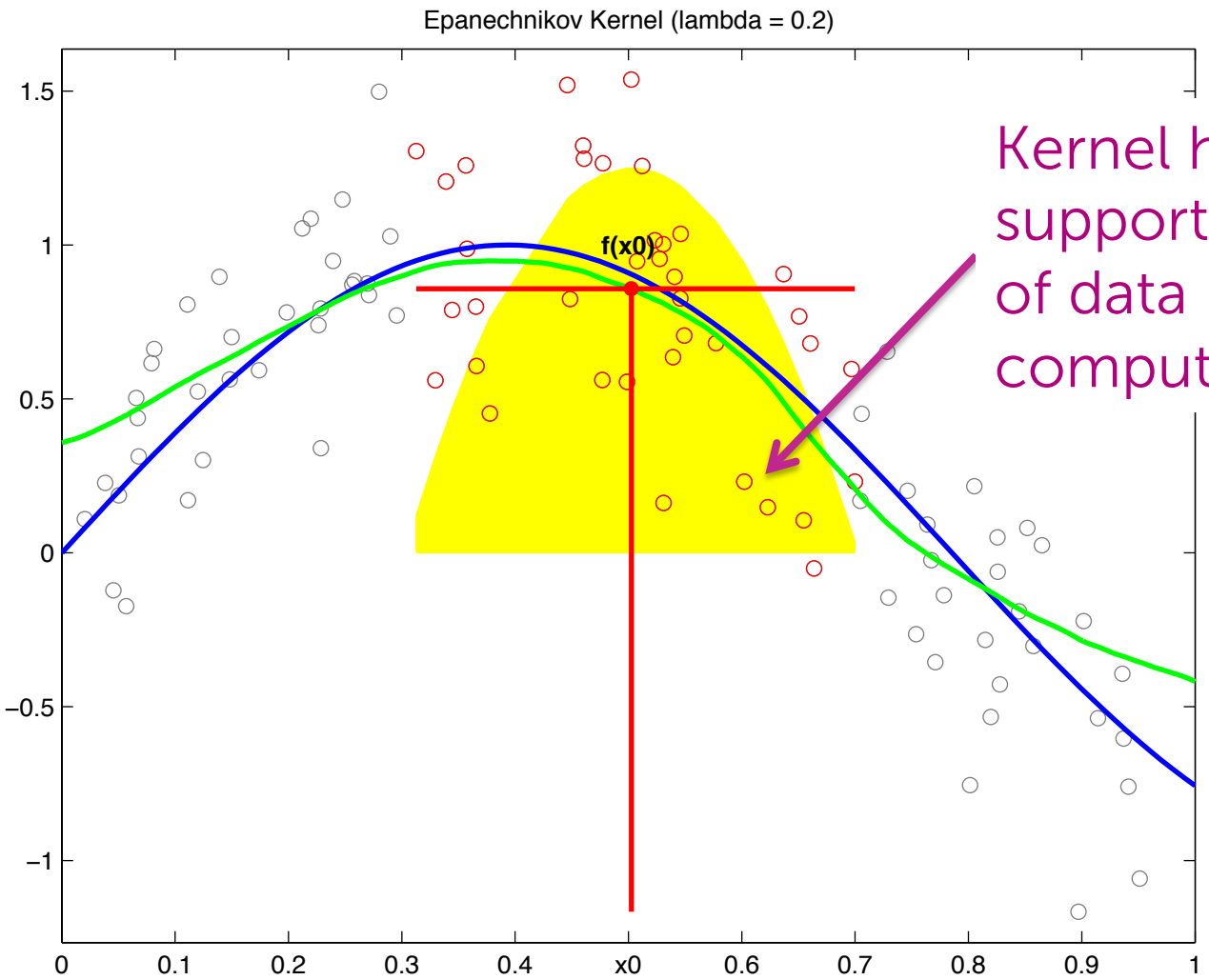
weight on each datapoint

$$\hat{y}_q = \frac{\sum_{i=1}^{N} c_{qi} y_i}{\sum_{i=1}^{N} c_{qi}} = \frac{\sum_{i=1}^{N} \text{Kernel}_\lambda(\text{distance}(\mathbf{x}_i, \mathbf{x}_q)) * y_i}{\sum_{i=1}^{N} \text{Kernel}_\lambda(\text{distance}(\mathbf{x}_i, \mathbf{x}_q))}$$

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Kernel regression in practice

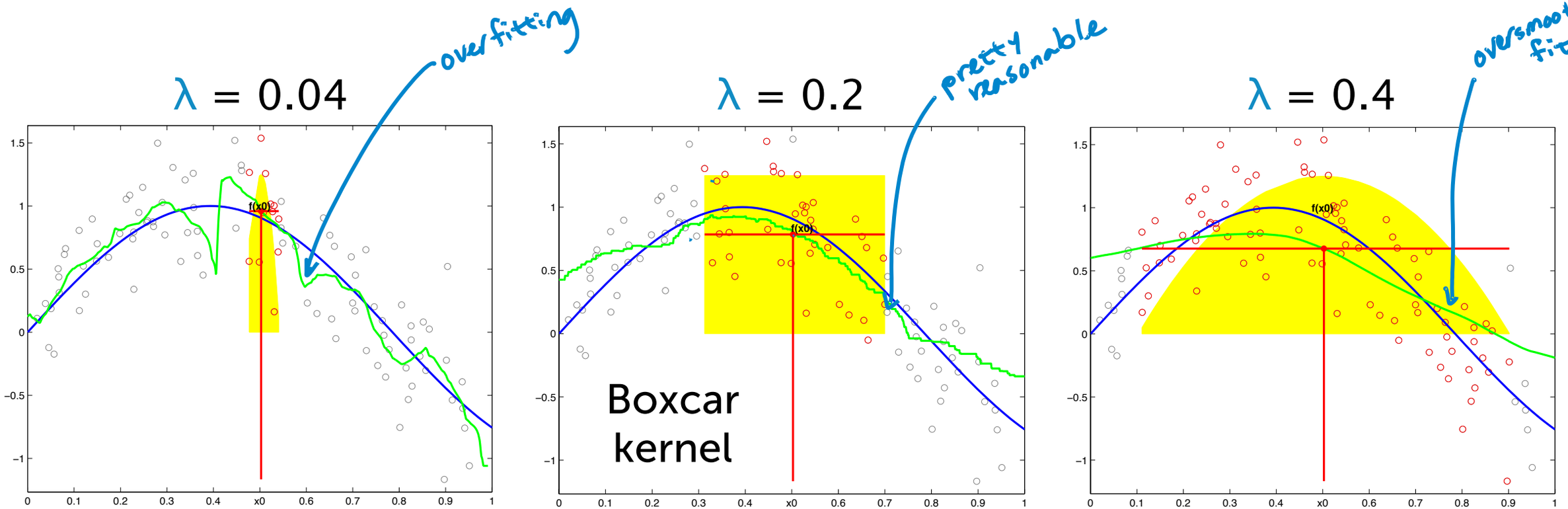


Kernel has bounded support…Only subset of data needed to compute local fit

©2015 Emily Fox & Carlos Guestrin

Choice of bandwidth λ

Often, choice of kernel matters much less than choice of λ



©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Choosing λ (or k in k-NN)

How to choose? Same story as always…

Cross Validation

©2015 Emily Fox & Carlos Guestrin

Formalizing the idea of local fits

©2015 Emily Fox & Carlos Guestrin

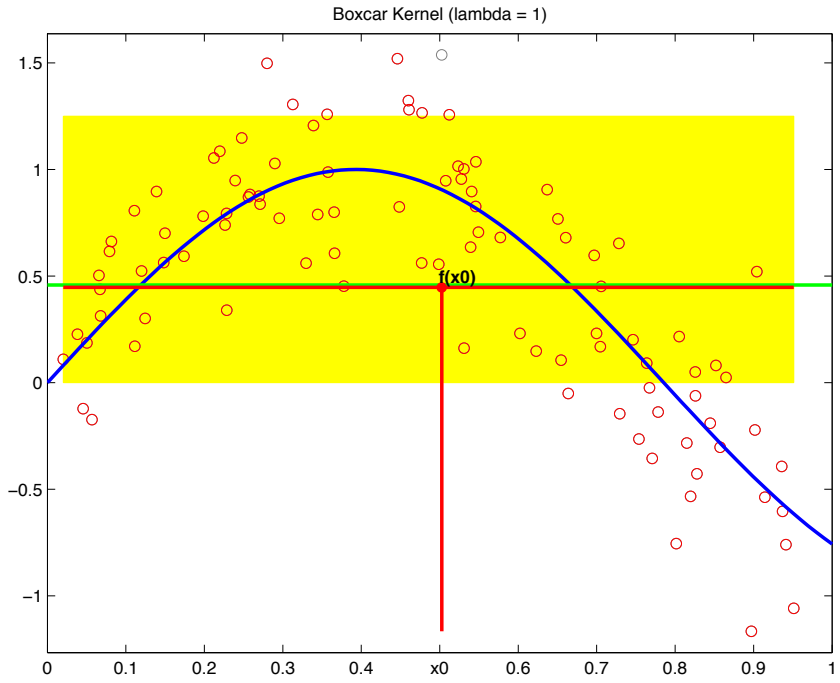
Troy University Resources

Contrasting with global average

A globally constant fit weights all points equally

$$\hat{y}_q = \frac{1}{N}\sum_{i=1}^{N} y_i = \frac{\sum_{i=1}^{N} c\, y_i}{\sum_{i=1}^{N} c}$$

equal weight on each datapoint



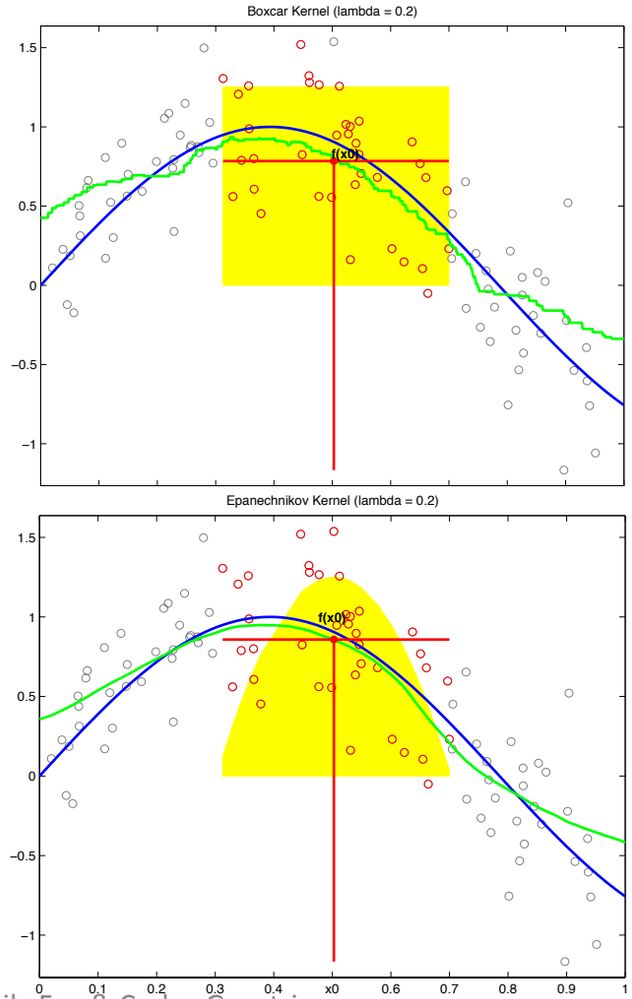
©2015 Emily Fox & Carlos Guestrin

Contrasting with global average

Kernel regression leads to locally constant fit

– slowly add in some points and and let others gradually die off

$$\hat{y}_q = \frac{\sum_{i=1}^{N} \text{Kernel}_\lambda(\text{distance}(\mathbf{x}_i,\mathbf{x}_q)) * y_i}{\sum_{i=1}^{N} \text{Kernel}_\lambda(\text{distance}(\mathbf{x}_i,\mathbf{x}_q))}$$



©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Local linear regression

So far, discussed fitting constant function locally at each point

→ "locally weighted averages"

Can instead fit a line or polynomial locally at each point

→ "locally weighted linear regression"

©2015 Emily Fox & Carlos Guestrin

Local regression rules of thumb

- Local linear fit reduces bias at boundaries with minimum increase in variance
- Local quadratic fit doesn't help at boundaries and increases variance, but does help capture curvature in the interior
- With sufficient data, local polynomials of odd degree dominate those of even degree

Recommended default choice:

local linear regression

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Discussion on k-NN and kernel regression

Nonparametric approaches

k-NN and kernel regression are examples of nonparametric regression

General goals of nonparametrics:

- Flexibility
- Make few assumptions about f(**x**)
- Complexity can grow with the number of observations N

Lots of other choices:

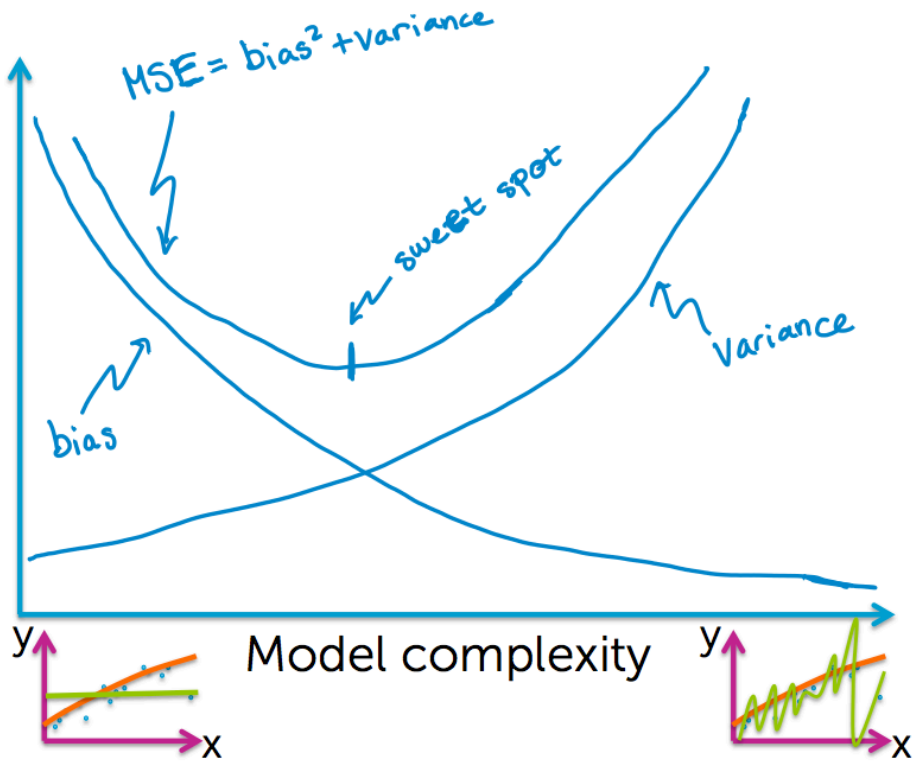
- Splines, trees, locally weighted structured regression models…

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Limiting behavior of NN: Noiseless setting (ε_i=0)

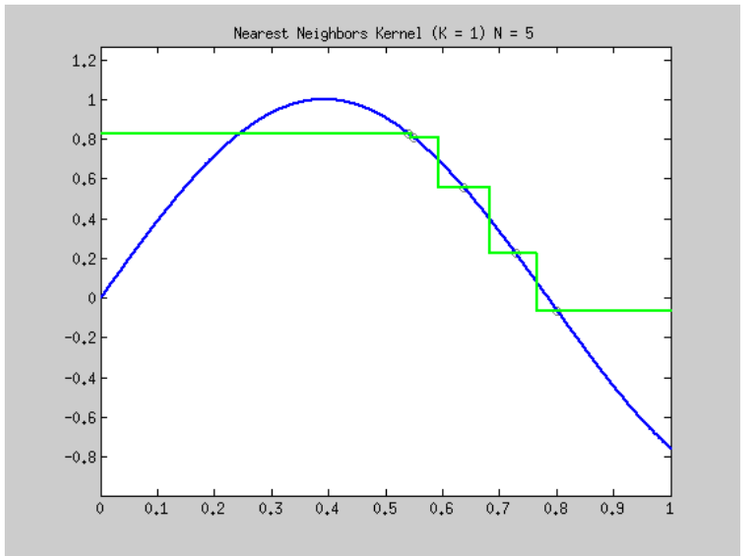
In the limit of getting an infinite amount of noiseless data, the MSE of 1-NN fit goes to 0



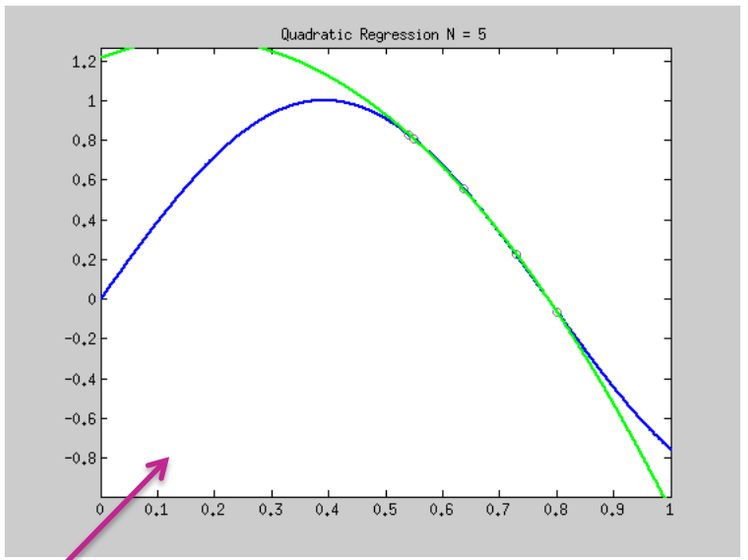
©2015 Emily Fox & Carlos Guestrin

Limiting behavior of NN: Noiseless setting (ε_i=0)

In the limit of getting an infinite amount of noiseless data, the MSE of 1-NN fit goes to 0



1-NN fit



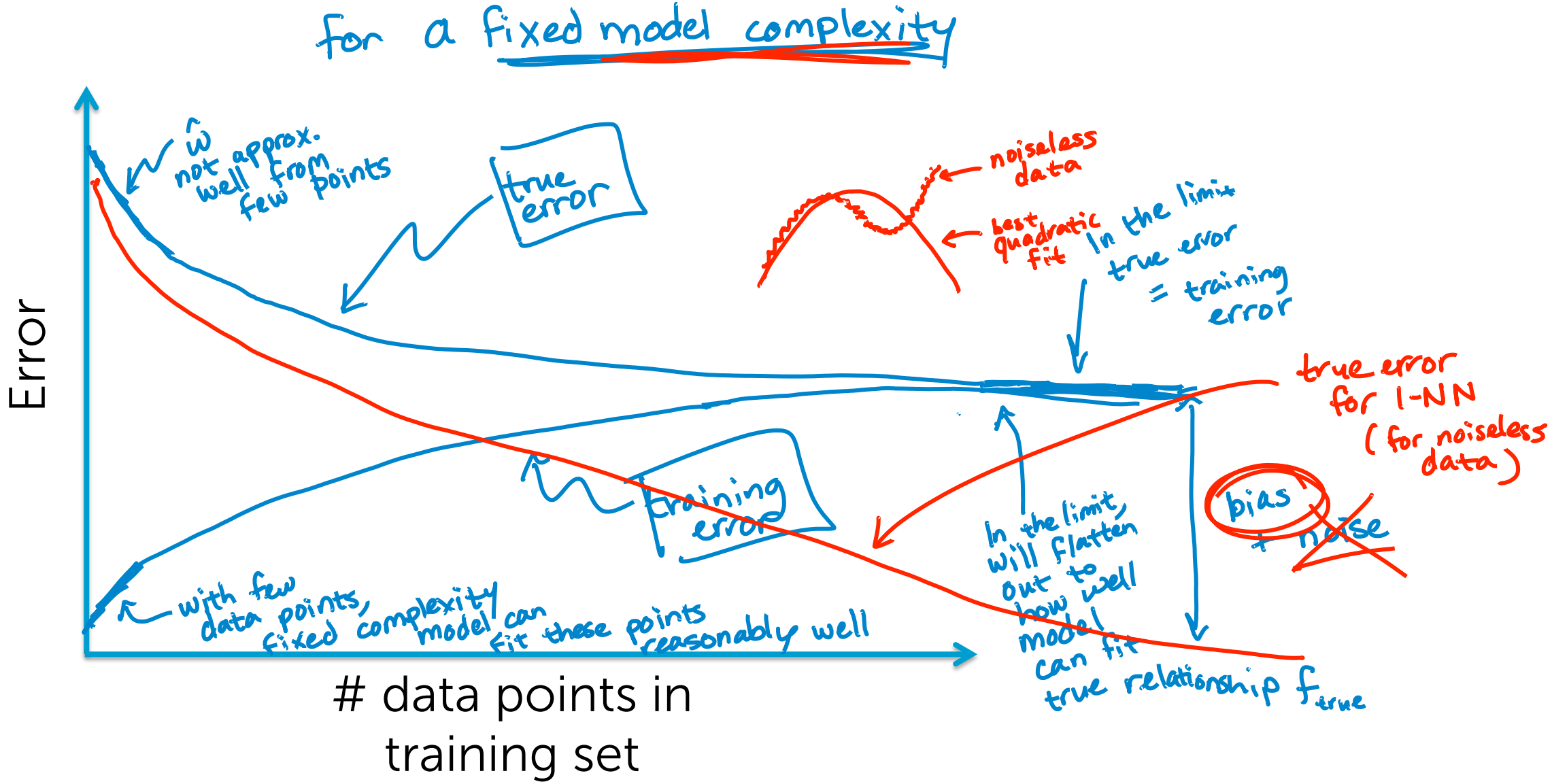
Quadratic fit

Not true for parametric models!

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

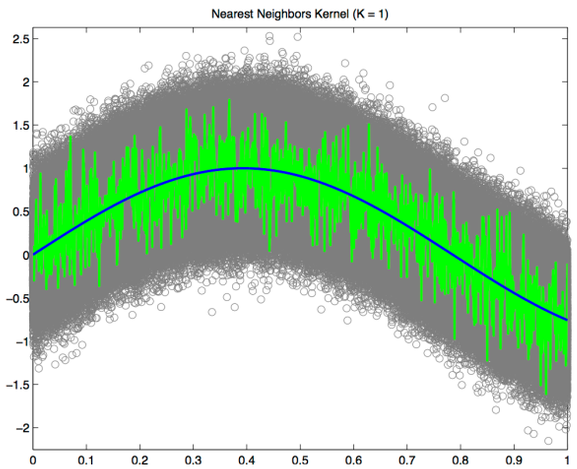
Error vs. amount of data



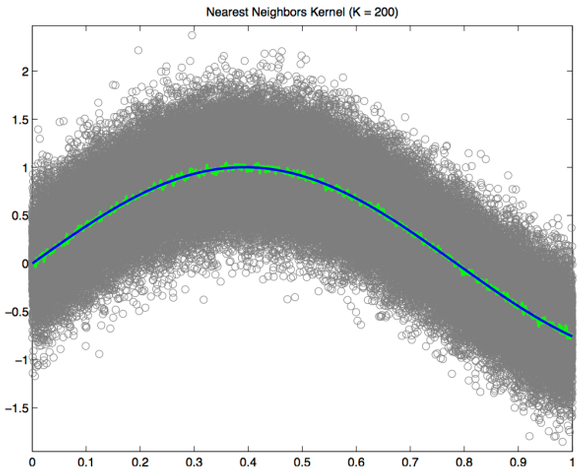
©2015 Emily Fox & Carlos Guestrin

Limiting behavior of NN: Noisy data setting

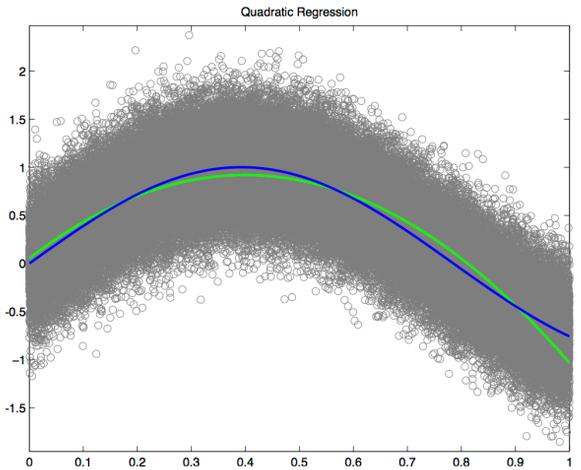
In the limit of getting an infinite amount of data, the MSE of NN fit goes to 0 if k grows, too



1-NN fit



200-NN fit



Quadratic fit

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

NN and kernel methods for large d or small N

NN and kernel methods work well when the data cover the space, but…

- the more dimensions d you have, the more points N you need to cover the space
- need N = O(exp(d)) data points for good performance

This is where parametric models become useful…

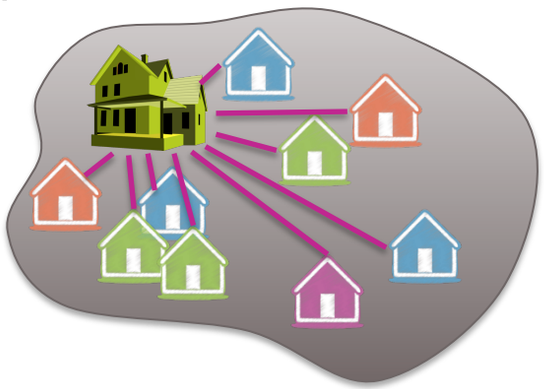
©2015 Emily Fox & Carlos Guestrin

Complexity of NN search

Naïve approach: **Brute force search**

- Given a query point $\mathbf{x}_q$
- Scan through each point $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N$
- O(N) distance computations per 1-NN query!
- O(Nlogk) per k-NN query!

What if N is huge???
(and many queries)



Will talk more about efficient methods in Clustering & Retrieval course

©2015 Emily Fox & Carlos Guestrin

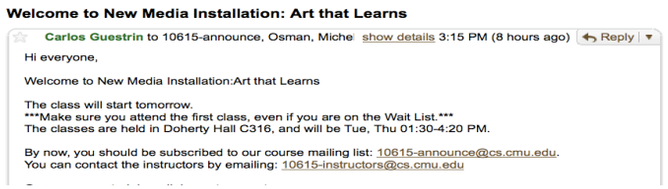
Troy University Resources

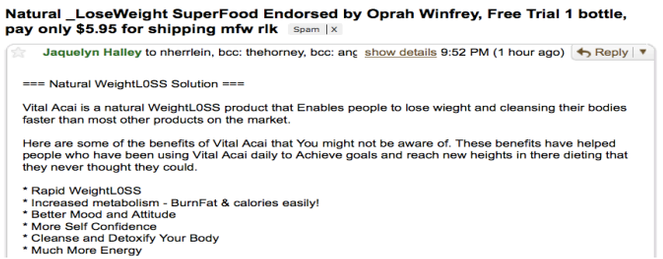
k-NN for classification

©2015 Emily Fox & Carlos Guestrin

Recall classification task: Spam filtering example







Not spam

Spam

Input: **x**

Output: y

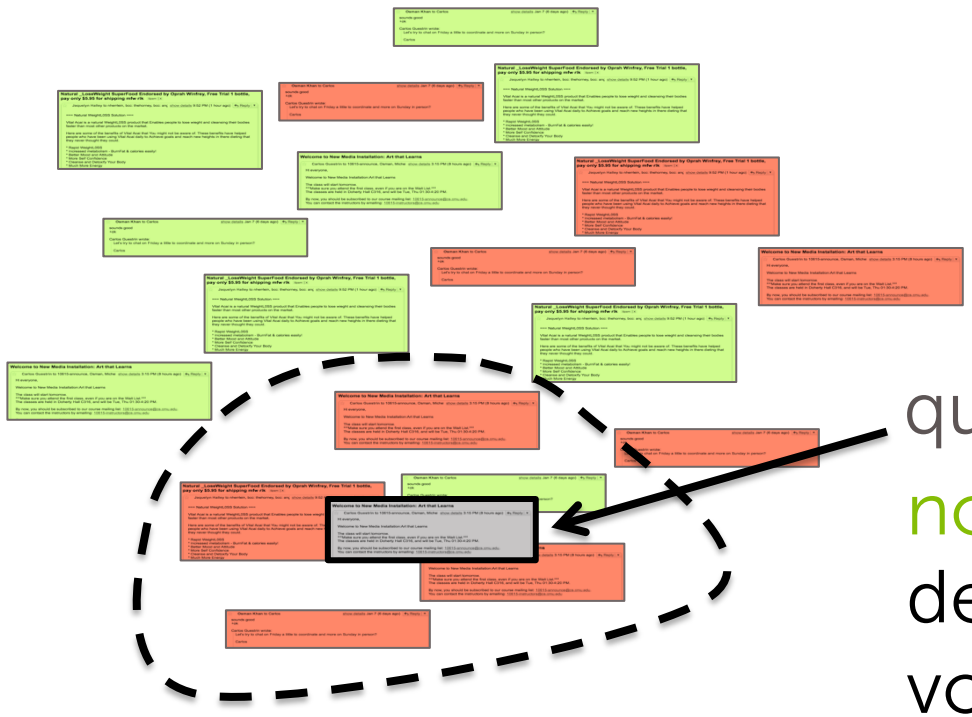
Text of email, sender, IP,…

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Using k-NN for classification

Space of labeled emails (not spam vs. spam), organized by similarity of text



query email

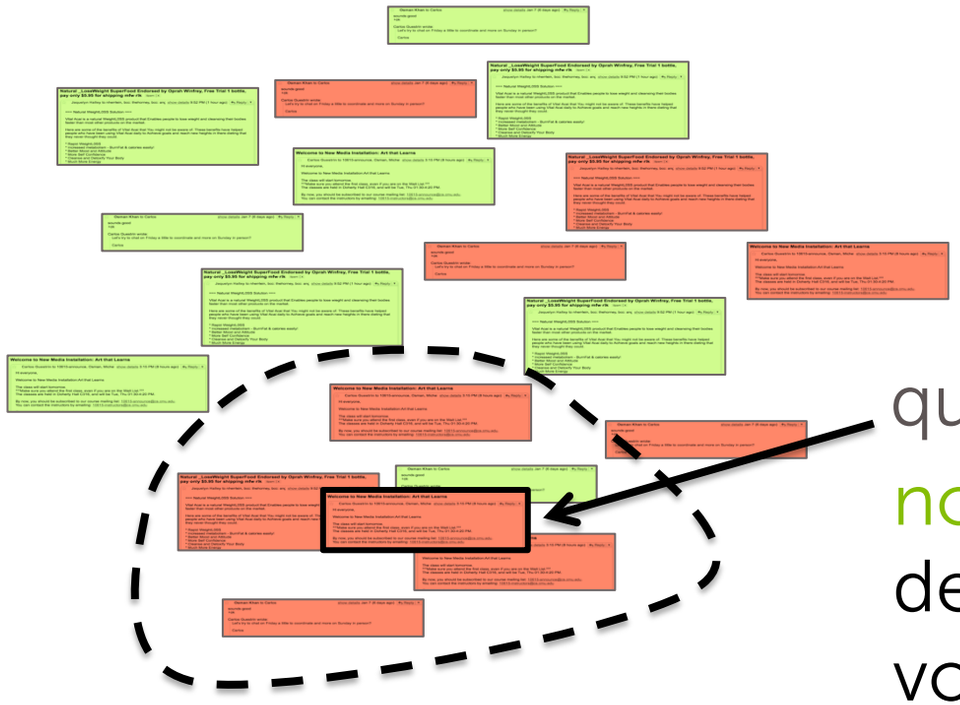
not spam vs. spam

decide via majority vote of k-NN

©2015 Emily Fox & Carlos Guestrin

Using k-NN for classification

Space of labeled emails (not spam vs. spam), organized by similarity of text



query email

not spam vs. spam

decide via majority vote of k-NN

©2015 Emily Fox & Carlos Guestrin

Troy University Resources

Summary for nearest neighbor and kernel regression

©2015 Emily Fox & Carlos Guestrin

What you can do now…

- Motivate the use of nearest neighbor (NN) regression
- Define distance metrics in 1D and multiple dimensions
- Perform NN and k-NN regression
- Analyze computational costs of these algorithms
- Discuss sensitivity of NN to lack of data, dimensionality, and noise
- Perform weighted k-NN and define weights using a kernel
- Define and implement kernel regression
- Describe the effect of varying the kernel bandwidth λ or # of nearest neighbors k
- Select λ or k using cross validation
- Compare and contrast kernel regression with a global average fit
- Define what makes an approach nonparametric and why NN and kernel regression are considered nonparametric methods
- Analyze the limiting behavior of NN regression
- Use NN for classification

©2015 Emily Fox & Carlos Guestrin

Troy University Resources